Projected Health Impact of Achieving WHO Targets for Cardiovascular Risk-Factor Control

Work in Progress

WHO · University of Washington

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Study Overview

Aim 1: Research Questions

Central Question

What is the projected **burden-of-disease impact** of achieving the WHO cardiovascular risk-factor control targets by **2030**?

Sub-questions driving the analysis:

1. How many **cardiovascular events** could be averted?
2. How many **CVD deaths** could be prevented by 2030 and beyond?
3. How would **life-years, YLLs, YLDs, and DALYs** change under joint achievement of:
   - 150M additional people with controlled hypertension
   - $\geq$50% lipid-lowering therapy coverage
   - 80% BP control among people with diabetes
   - 80% validated BP device availability

Scope

190 countries · Ages 20–95 · 2026–2050 · Multiple CVD causes · WHO Target Scenarios

Intervention start: 2026

Model calibrated to 2019 baseline. First year of target scale-up effects entering the projection is **2026**.

Study Design at a Glance

Component	Detail
Model type	Discrete-time multi-state model (Well $\rightarrow$ Sick $\rightarrow$ Dead)
Calibration data	GBD 2023: incidence, case fatality, prevalence; all-cause & CVD-specific
Population data	UNWPP 2024 single-year age/sex projections
Intervention start	**2026** — first year intervention effects enter the projection
Scale-up period	Linear scale-up 2026 $\rightarrow$ 2030; rates held constant 2030–2050
Comparison	Business-as-usual: 2024 control rates, no additional scale-up
Outcome	CVD deaths averted vs. BAU; age-standardised mortality rate (ASMR)
Time horizon	2026–2050 (25-year horizon); model calibrated from 2019

Study Context

Objective

- Quantify CVD deaths avertable by achieving WHO 2030 cardiovascular targets
- Coverage: **190 countries**, 2025–2050
- Four causes: IHD, ischemic stroke, hemorrhagic stroke, hypertensive HD

Model

- Discrete-time state transition (Well $\rightarrow$ Sick $\rightarrow$ Dead)
- Calibrated to GBD 2023; projected with UNWPP 2024 populations
- 8 blood-pressure bins
- Baseline treatment effects explicitly de-duplicated

Research Question

What are the projected deaths averted from achieving WHO hypertension control and lipid-lowering therapy targets across **190 countries**, 2026–2050?

Pending △

Confirm primary metric: deaths averted vs. DALYs vs. LYG

Data Sources

Data Sources and Key Variables

Source	Variables extracted	Role in model
GBD 2023 (IHME)	IR, CF, prevalence, background mortality (age/sex/country)	Baseline transition rates; initial states
GBD 2019 (IHME)	Relative risks per 10 mmHg SBP increase (IHD, stroke, HHD, by age)	Anchor BP-bin incidence; normalisation factor α
Ettehad et al. 2016	Trial effect sizes per BP category; diabetes-stratified	Treatment effect sizes by BP bin $\times$ cause
UNWPP 2024 (UN)	Population projections 2019–2050 (single-year age, sex, country)	Cohort sizes; rate denominators
NCD-RisC	Baseline HTN control rates (C_0); BP distribution by country	C_0; $P(\text{BP}_{\text{cat}})$
Country-specific	HTN control targets 2030 (Progress / Ambitious / Aspirational)	C_{target} per scenario

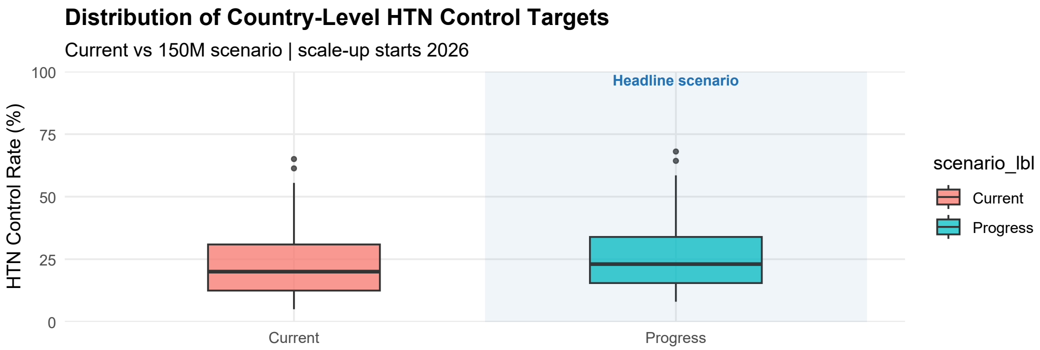
Scenarios

Four WHO Priority Targets

#	WHO Target	Modelled Scenario
1	150M more with HTN controlled by 2030	**HTN Control (General)**: $\approx$ 100M more people with BP control by 2030
2	$\geq$50% eligible adults on lipid-lowering therapy by 2030	**Improved Statin Uptake**: 50% coverage by 2030
3	80% BP control among diagnosed diabetics by 2030	**HTN Control (Diabetes)**: 80% BP control in diabetics by 2030 ($\approx$ 50M more people with BP control)
4	80% availability of validated BP devices	Operationally embedded in HTN scenarios (150M Target [1+3])

Combination rule: $\varepsilon_{IR,\text{total}} = \varepsilon_{IR,\text{BP}} \times \varepsilon_{IR,\text{statin}}$

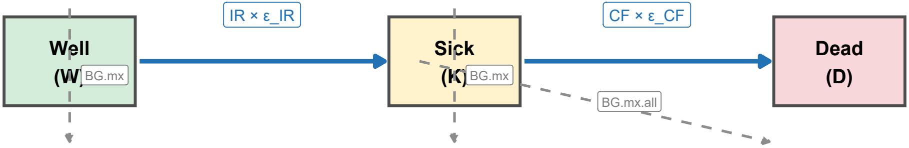
HTN Control Targets by 2030 — Country Distribution



Mathematical Model

Model Architecture: Three Health States

Multi-State Transition Model | Annual time step | Ages 20-95 | 4 CVD causes



Key: $\varepsilon_{IR}, \varepsilon_{CF}$ = multiplicative effect ratios from the intervention module · BG.mx = background mortality · COVID-19 excess mortality applied 2020–2021 only

Modeling Assumptions

Model structure

- Discrete-time annual step (2019–2050); ages 20–95
- Four causes modeled independently
- COVID-19 excess mortality applied 2020–2021 only; no long-term effects

Intervention mechanics

- Linear scale-up 2026 $\rightarrow$ 2030; held constant 2030–2050
- Only incremental coverage above C_0 (or $U_{0,\ell}$) generates new effects (no double-counting)
- Full adherence assumed for HTN control, statins adherence adjusted down for primary and secondary prevention.
- Statin effects restricted to ages ≥ 40 and IHD & ischemic stroke only

Blood pressure & effect sizes

- Static log-normal BP distribution: no secular trend, no endogenous updating as coverage rises
- Treatment eligibility: $\geq$140 mmHg (5 hypertensive bins)
- GBD RRs anchor bin-level baseline incidence
- Diabetes subgroup: base control rates assumed 15 percentage points higher than general HTN
- Diabetes subgroup: multiplicative reduction blended with general HTN effects using diabetes prevalence and high blood pressure prevalence as weights

Combination of interventions

- Effects combine **multiplicatively** across all interventions
- BP-related effects combined first, then with statins
- No saturation, synergy, or interaction terms beyond multiplicative combination

Burden of Disease: Methods

Three burden-of-disease metrics

- **YLL** (Years of Life Lost): deaths $\times$ remaining life expectancy at age of death
- **YLD** (Years Lived with Disability): prevalent CVD cases $\times$ disability weight
- **DALY** (Disability-Adjusted Life Years): $DALY = YLL + YLD$

DALYs averted = $DALY_{\text{BAU}} - DALY_{\text{intervention}}$

Full derivation in Supplemental Mathematical Details.

Reference standards

Life table: UNWPP 2024, medium variant (country- and year-specific e_x)

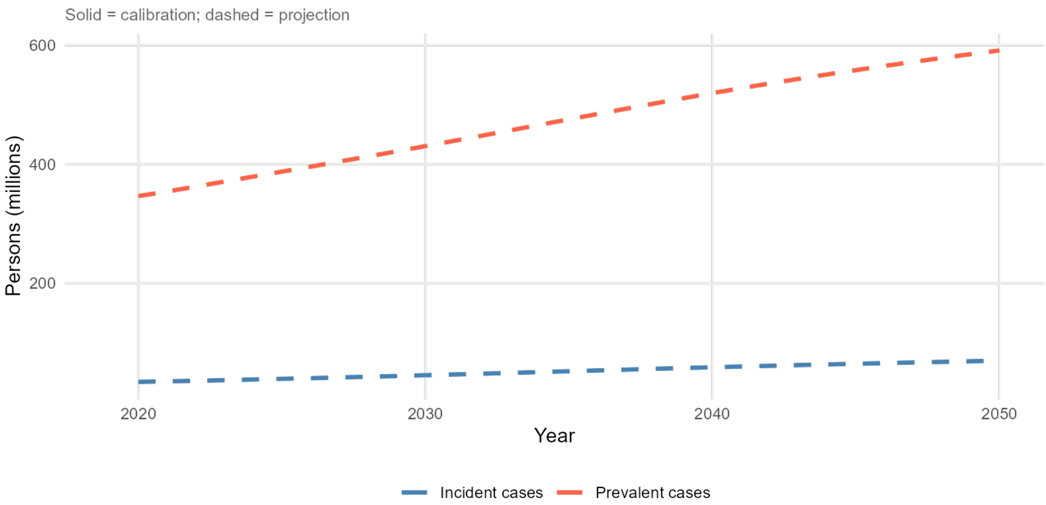
Disability weights: GBD 2023, derived as YLD/Prevalence per cause and location

Discounting: None (undiscounted)

Age weighting: None

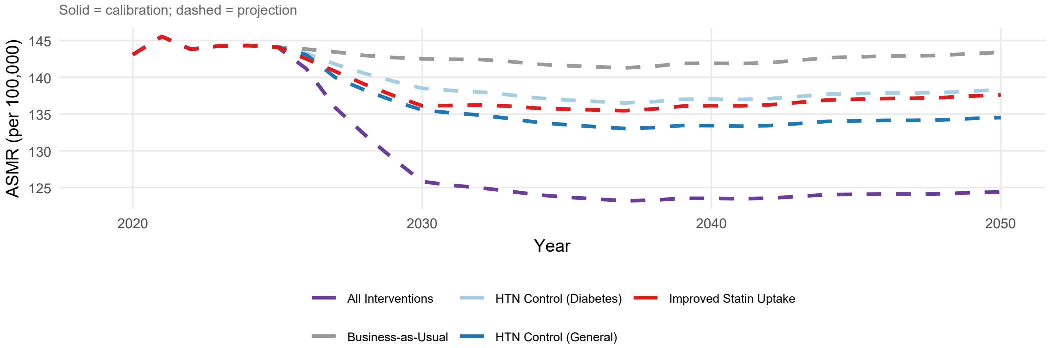
Results

Baseline Disease Trends (Business-as-Usual)



Incident and prevalent CVD cases 2019–2050; solid = calibration, dashed = projection.

Age-Standardised CVD Mortality Rate



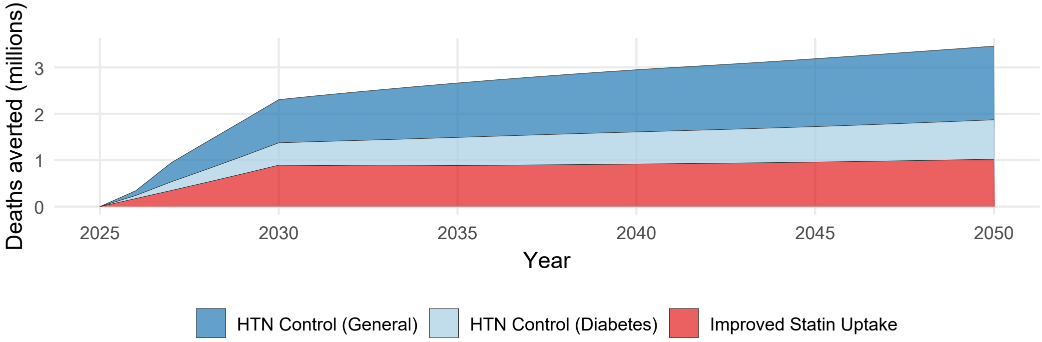
Summary: Cumulative Deaths Averted, 2026–2050

Scenario	Cumulative (M)	Avg./year (M)	% of BAU deaths
HTN Control (General)	29.4 M	1.2 M	4.2%
HTN Control (Diabetes)	15.3 M	0.6 M	2.2%
Improved Statin Uptake	21.3 M	0.9 M	3.1%
All Interventions	**64.7 M**	**2.6 M**	**9.3%**

Headline

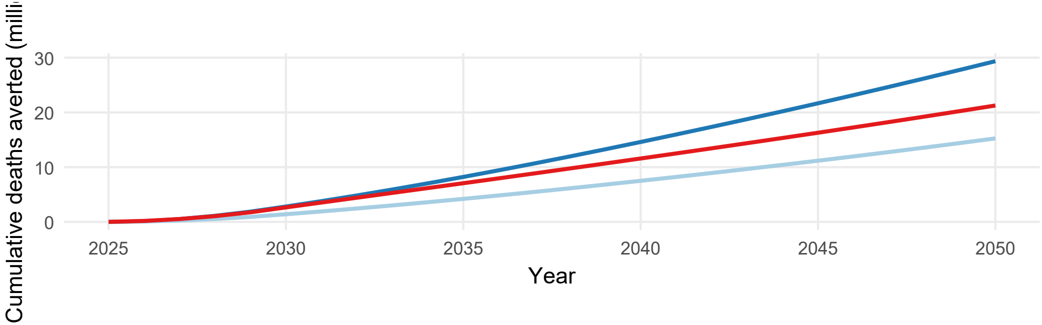
All Interventions: **64.7 million deaths averted** globally, 2026–2050 ($\approx$ 2.6 M/year).

Annual Deaths Averted Over Time



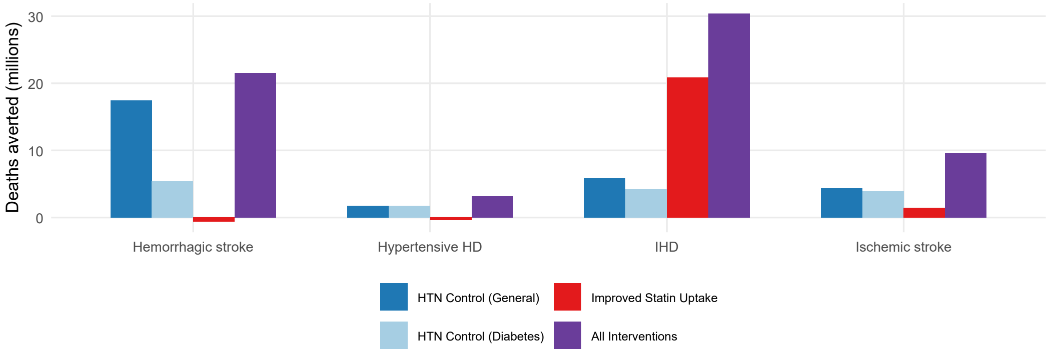
Cumulative deaths averted per intervention relative to business-as-usual.

Cumulative Deaths Averted Over Time

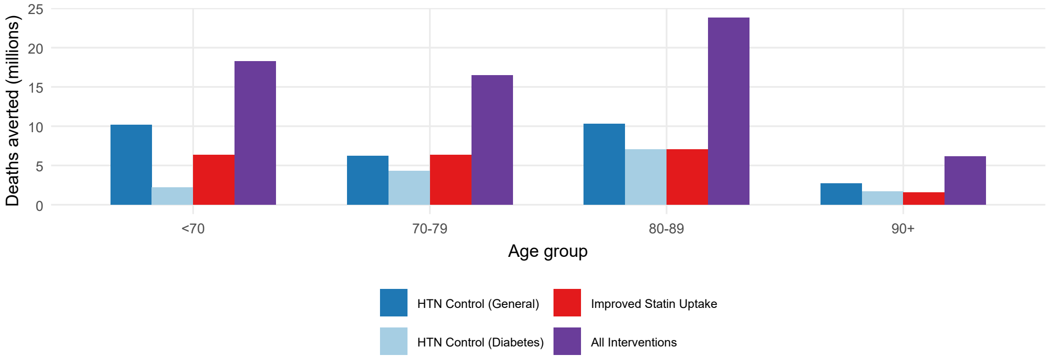


Cumulative deaths averted per intervention relative to business-as-usual.

Deaths Averted by Cause of Death



Deaths Averted by Age Group



Burden of Disease Averted: YLLs, YLDs, and DALYs

Combined scenario (All Interventions), 2026–2050

- **DALYs averted:** `[requires sl_scalars$total_dalys_all]`
- **YLLs averted:** `[requires sl_scalars$total_ylls_all]`
- **YLDs averted:** `[requires sl_scalars$total_ylds_all]`

By individual intervention

- HTN Control (General): `[requires sl_scalars$total_dalys_bp]` DALYs
- HTN Control (Diabetes): `[requires sl_scalars$total_dalys_bpd]` DALYs
- Improved Statin Uptake: `[requires sl_scalars$total_dalys_stat]` DALYs

YLL vs. YLD

YLLs (premature mortality) dominate the DALY total; YLDs (morbidity from prevalent cases) contribute a smaller but non-negligible share, reflecting the large prevalent CVD pool.

See Table 3 in the manuscript for the full burden-of-disease summary by intervention scenario.

Regional Summary

Region	BAU deaths (M)	Averted (M)	% averted
Western Pacific	227.3	21.5	9.4%
South-East Asia	175.8	17.0	9.6%
Europe	117.2	9.6	8.2%
Africa	48.7	5.8	12%
Eastern Mediterranean	64.3	5.7	8.9%
Americas	62.8	5.1	8.1%

Observation

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

Discussion

Interpretation

Key findings

- **All Interventions** yields largest gains: HTN control + statins are complementary
- **IHD** and **ischemic stroke** drive the majority of avertable deaths
- Gains **accelerate after 2030** as coverage compounds
- **Older adults (70+)** benefit most in absolute terms
- **Top 15 countries** account for $>$75% of avertable burden

Policy Implication

Combined WHO Targets 1 + 2 are synergistic — neither alone captures the full benefit.

Uncertainty △

Sensitivity analysis to diabetes targeting assumptions and secular trends pending.

Limitations

Model structure & assumptions

- Stable transition rates assumed; no dynamic feedback between states
- Static BP distribution (log-normal): no secular trend and no endogenous updating as coverage increases
- No heterogeneity in adherence: newly HTN covered individuals assumed fully adherent
- COVID-19 excess mortality only modelled for 2020–2021; long-term effects excluded

Data & parameters

- RCT-based effect sizes (e.g., Ettehad ρ_{CF}): real-world effectiveness at scale may be lower
- GBD 2023 uncertainty intervals not propagated
- Baseline C_0 from NCD-RisC; measurement error unquantified
- BP distributions rely on GBD modelling (not direct surveys in all countries)

Scenarios & implementation realism

- Linear scale-up assumed (2026–2030); real programmes may follow non-linear trajectories
- No explicit health-system capacity constraints modelled
- Interactions with statins, diabetes care, and BP device availability not included

Next Steps

Analysis priorities

1. Sensitivity analysis on effect sizes, trends and coverage
2. Add life-years gained (LYG) and DALYs as co-primary outcome
3. Stratify by income group (World Bank)
4. Conduct Aim 2: simulate alternative scale-up trajectories and implementation scenarios for hypertension control

Pending Decisions △

- Journal target?

Summary

1. The gains are large and achievable

All-interventions: **64.7 million deaths averted** globally, 2026–2050.

2. HTN control + statins compound risk reduction

HTN control and statins have multiplicative risk reductions, producing larger combined effects.

3. Regional patterns reflect burden and coverage gaps

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

Meeting WHO 2030 CVD targets is not just aspirational — it is quantifiably lifesaving.

Appendix

Notation: Core Model

Symbol	Description
Core state-transition model	
a, s, c, t, ℓ	Age, sex, cause, year, location
W, K, D	Well, Sick, Dead populations
$IR_{a,s,c}$, $CF_{a,s,c}$	Baseline incidence & case-fatality rates
$\varepsilon_{IR,j}$, $\varepsilon_{CF,j}$	Multiplicative effect ratios for intervention j
$BG.mx$, $BG.mx.all$	Cause-specific & all-cause background mortality
$covid.mx$	COVID-19 mortality (2020–2021 only)

Notation: HTN Control

Symbol	Description
HTN control (general & diabetes subgroup)	
$C_0, C_t, C_{\text{target}}$	Baseline, time-t, and target control rates
$\Delta C_{\text{target}}, \Delta C_{\text{increment}}$	Total & time-varying incremental coverage
C^{Δ}_{agg}	Population-weighted incremental coverage (hypertensive bins)
$P(\text{BP}_{\text{cat}})$	BP category population distribution (8 bins)
RR_{GBD}	GBD relative risk per 10 mmHg SBP increase
α_{inc}	Incidence normalization factor
$E_{\text{trial}}, E_{\text{DM}}, E_{\text{noDM}}$	Ettehad effect sizes (blended, DM-only, non-DM)
w_{DM}	Diabetes prevalence weight (general HTN: 0.10; DM subgroup: 1.0)
$\rho_{CF,c}$	CF reduction factor per unit incremental coverage
$T_{\text{start}}, T_{\text{end}}$	Scale-up start (2026) & end (2030)

Notation: Statins & Combined Effects

Symbol	Description
Statin therapy	
$U_{0,\ell}$, U_{target}	Baseline & target statin coverage
$\Delta U(t,\ell)$	Incremental statin coverage above $U_{0,\ell}$ at time t
$T_{\text{stat,start}}$, $T_{\text{stat,end}}$	Statin scale-up start (2026) & end years
RR_c^{IR}, RR_c^{CF}	Trial-based RRs for incidence & case fatality
δ_c^{IR}, δ_c^{CF}	Risk reductions $= 1 - RR$
$AF_{c,\ell}$	Cause- & location-specific attributable fraction (incidence only)
ψ_{athero}	Atherosclerotic fraction of ischemic stroke ($= 0.60$)
$E_{c,\ell}^{IR}(t)$, $E_{c,\ell}^{CF}(t)$	Incremental effect sizes for IR and CF

HTN Intervention Module: 9-Step Pipeline

	Action	Output
1	Baseline BP distribution $P(\text{BP}_{\text{cat}})$ without treatment (rx $= 0$)	$P(\textit{BP}_{\textit{cat}} \mid a, s, t)$
2	GBD relative risks per 10 mmHg SBP $\rightarrow$ bin-specific incidence IR_{bin}	$IR_{\textit{bin}}$
3	Diabetes-weighted Ettehad trial effect sizes E_{trial} per BP bin $\times$ cause	$E_{\textit{trial}}$
4	Linear coverage scale-up from C_0 to C_{target}, **starting 2026**	$C_t(t,\ \textit{BP}_{\textit{cat}})$
5	Coverage adjustment: $E_{\text{adj}} = E_{\text{trial}}(C_t - C_0)\,/\,(1 - E_{\text{trial}}\,C_0)$	$E_{\textit{adj}}(t)$
6	New bin incidence: $IR_{\text{bin,new}} = IR_{\text{bin}} \times (1 - E_{\text{adj}})$	$IR_{\textit{bin,new}}$
7	Aggregate: $IR_{\text{new}} = \sum IR_{\text{bin,new}} \times P(\text{BP}_{\text{cat}})$	$IR_{\textit{new}}$
8	Incidence effect ratio: $\varepsilon_{IR} = IR_{\text{new}}\,/\,IR_{\text{original}}$	$\varepsilon_{IR} \rightarrow$ *model*
9	Case-fatality effect ratio: $\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$	$\varepsilon_{CF} \rightarrow$ *model*

Blue row = intervention start (2026). Green rows = outputs passed to state-transition model.

HTN Control: Coverage Scale-Up and Key Equations

Incidence pathway

Step 2 — GBD RR anchors bin incidence:

$$IR_{\text{bin}} = \frac{RR_{\text{GBD}} \times IR}{\alpha}, \quad \alpha = \sum_{\text{cat}} P_{\text{cat}} \cdot RR_{\text{GBD}}$$

Step 5 — baseline-adjusted effect (no double-counting):

$$E_{\text{adj}}(t) = \frac{E_{\text{trial}}\,(C_t - C_0)}{1 - E_{\text{trial}}\,C_0}$$

Steps 7–8 — population-weighted effect ratio:

$$\varepsilon_{IR} = \frac{\sum IR_{\text{bin,new}} \cdot P_{\text{cat}}}{IR_{\text{original}}}$$

Case fatality pathway

Incremental coverage above baseline:

$$\Delta C_\delta(t) = \max(C_t - C_0,\ 0)$$

Aggregate across hypertensive bins (≥140 mmHg):

$$C^{\Delta}_{\text{agg}} = \frac{\sum_{\geq 140} \Delta C_\delta \cdot P_{\text{cat}}}{\sum_{\geq 140} P_{\text{cat}}}$$

CF effect ratio:

$$\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$$

Cause	ρ_{CF}
IHD	0.24
Ischemic stroke	0.36
Hemorrhagic stroke	0.76
HHD	0.20

HTN Control (Diabetes Subgroup)

WHO Target: 80% BP control among diagnosed diabetics by 2030

Same 9-step pipeline as general HTN. Key distinction: diabetes-specific Ettehad effect sizes ($w_{\text{DM}} = 1$):

General HTN (DM-blended):

$$E_{\text{trial}} = (1 - w_{\text{DM}})\, E_{\text{noDM}} + w_{\text{DM}}\, E_{\text{DM}}, \quad w_{\text{DM}} = 0.10$$

Diabetes subgroup (DM-only):

$$E^{\text{DM}}_{\text{trial}} = E_{\text{DM}}$$

Coverage scale-up to $C^{DM}_{target} = 0.80$:

$$C^{\text{DM}}_t = C^{\text{DM}}_0 + \Delta C^{\text{DM}} \ \min\left[\frac{t - 2026}{4},\ 1\right]$$

Effect ratios (same structure as general HTN):

$$\varepsilon_{IR,\text{DM}}(a, s, c, t) = \frac{IR^{\text{DM}}_{\text{new}}}{IR_{\text{original}}}$$

$$\varepsilon_{CF,\text{DM}} = 1 - \rho_{CF,c} \times C^{\Delta,\text{DM}}_{\text{agg}}$$

Why a separate module?

People with diabetes have higher CVD risk at any BP level. DM-specific Ettehad effects are larger (especially for stroke) than blended effects at $w_{\text{DM}} = 0.10$.

When combined with general HTN

Effects multiply (no double-counting via incremental coverage formulation):

$$\varepsilon_{IR,\text{BP,total}} = \varepsilon_{IR,\text{gen}} \times \varepsilon_{IR,\text{DM}}$$

Statin Therapy: Coverage Scale-Up and Key Equations

Scope: IHD & ischemic stroke; ages ≥ 40 only

Incremental coverage above baseline $U_{0,\ell}$:

$$\Delta U(t,\ell) = \Delta U_{\text{target},\ell}\ \text{min}\left[\frac{t - T_{\text{stat,start}} + 1}{T_{\text{stat,end}} - T_{\text{stat,start}} + 1}, 1\right]$$

where $\Delta U_{\text{target},\ell} = \text{max}(U_{\text{target}} - U_{0,\ell},\ 0)$

Trial-based RRs (per ≈1 mmol/L LDL-C reduction):

Cause	RR^{IR}	RR^{CF}
IHD	0.74	0.80
Ischemic stroke	0.80	0.96

Incidence (primary prevention, AF-scaled):

$$E^{IR}_{c,\ell}(t) = \frac{AF_{c,\ell}\ \delta^{IR}_c\ \Delta U(t,\ell)}{1 - \delta^{IR}_c\ U_{0,\ell}}$$

Case fatality (secondary prevention; no AF; atherosclerotic restriction for stroke):

$$E^{CF}_{c,\ell}(t) = \begin{cases} \dfrac{\delta^{CF}_{\text{ihd}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ihd}}\ U_{0,\ell}}, & c = \text{ihd} \\[2ex] \dfrac{\psi_{\text{athero}}\ \delta^{CF}_{\text{ist}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ist}}\ U_{0,\ell}}, & c = \text{istroke} \end{cases}$$

$\delta^{IR}_c = 1 - RR^{IR}_c;\quad \psi_{\text{athero}} = 0.60$
(atherosclerotic fraction)

Effect ratios passed to model:

$$\varepsilon_{IR,\text{stat}} = \begin{cases} 1 - E^{IR}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,ist} \\ 1, & \text{otherwise} \end{cases}$$

$$\varepsilon_{CF,\text{stat}} = \begin{cases} 1 - E^{CF}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,is} \\ 1, & \text{otherwise} \end{cases}$$

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Thank You

Questions & Discussion

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